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Connecting and Controlling a Motorized Ball Valve

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Connecting and Controlling a Motorized Ball Valve

This tutorial will give you an overview of the core features of the board, setup the development environment and introduce the required APIs to program the board.

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Overview

A ball valve is a form of quarter-turn **valve** which uses a hollow, perforated and pivoting ball to control flow of liquids and gasses through it. This tutorial will guide you through connecting the board to a 3 Wire-Valve and writing a sketch that controls the basic operations such as the opening and closing of the valves.

Tip : If this is for your first Edge Control project, we recommend you to take a



look at the **User Manual of the product** to set up the development environment before you proceed.

Help

Goals

How to connect a motorized valve to the Edge Control board

How to control the valve through basic commands provided by the `Arduino_EdgeControl` library

How to power the board with an external power supply

Required Hardware and Software

Arduino Edge control board

1x **US Solid Motorized Ball Valve (9 - 24 V)** (or compatible)

External power source: 12V battery (LiPo / SLA) or power supply

1x Micro USB cable

Arduino IDE 1.8.10+

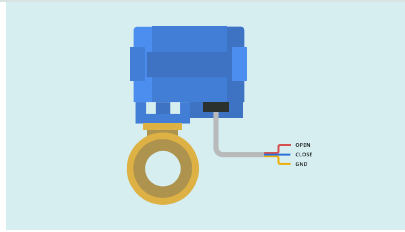
2x Phoenix Connectors

2x Jumper wires

Instructions

1. Connecting the Valves

The motorized valve comes with three wires primarily marked as blue, yellow and red. The red and blue cables are for the positive and negative signals and the yellow is for the ground.

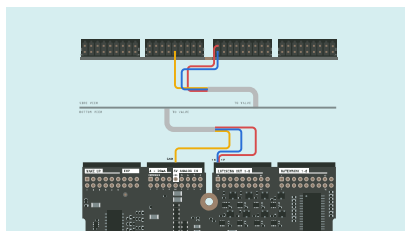


Before you attach the wires to the board you need to ensure that the Phoenix connectors are in place.

The Latching Circuitry A

Latching Circuit has 2 inputs for Positive(P) and Negative(N) signals and 1 output. Latches allow you to store the previous state of the signal that passes through either of the pins and maintains the state until you pass a new signal. Signal passed through the positive and the negative pins will also determine the direction of the signal to the valve. The Edge Control board comes with 8 pairs of Latches each labeled from 1N:1P up to 8N:8P.

Connect the red and the blue wire to **1N** and **1P** of your Edge Control board. As the valve doesn't come with internal drivers to store the state of the motor, we will use the **Latching_out** pins, instead of **Latching_out_cmd**, these are the ones that include drivers on the Edge Control.



Connect the yellow wire to the nearby `GND` pin. Ensure that the wires are fastened securely and tightly to the connectors so that they make contact with the pins.

2. Opening And Closing the Valves

Open a new sketch file on the Arduino IDE and name it `ValveControl.ino`. Add the header file `Arduino_EdgeControl.h` to your sketch.

```
1 #include <Arduino_EdgeC
```

Inside `void setup()`, after enabling the serial communication, run the initialization routine `EdgeControl.begin()`. This routine is in charge of enabling the default power areas of the board. Then use `Latching.begin()` to configure the expander pins as outputs.

```

1 void setup(){
2   Serial.begin(9600)
3   while(!Serial);
4
5   delay(1000);
6
7   Serial.println("3-
8
9   EdgeControl.begin(
10  Latching.begin();
11
12  Serial.println("St
13 }
```

As mentioned earlier, the 2 Input Pins of the latching circuit are primarily used to control the direction of the output signal. The

`<Arduino_EdgeControl.h>`

provides various methods to access and control these pins.

The command

`Latching.channelDirection(LA
TCHING_OUT_1, direction)`

is a method that can be used to assign a signal direction to a specific pin on the board through the parameters.

Inside the `loop()` you will add the instructions to open and close the valve.

`Latching.channelDirection()`

with the parameter

`LATCHING_OUT_1` to access the

1N:1P pins and the parameters,

`POSITIVE` or `NEGATIVE` for the

direction. If you want the valve to open you can use the function as such:

```

1 Latching.channelDirecti
```

And to close the valve, you need to send a signal in the opposite direction using the command:

```
1 Latching.channelDirecti
```

As it takes a few seconds for the valve to fully open or close, you need to maintain the signal for a set amount of time. Using the command,

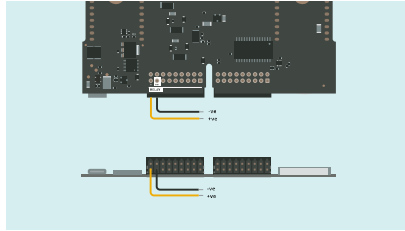
`Latching.strobe(4500)` you can adjust the duration (milliseconds) of a signal passing through a particular pin.

```
1 void loop(){
2   Serial.println("Cl
3   Latching.channelDi
4   Latching.strobe(45
5   delay(2500);
6
7   Serial.println("Op
8   Latching.channelDi
9   Latching.strobe(45
10  delay(2500);
11 }
```

3. Connecting To A Power Source

The valves require a power supply of 9 - 12 V and you can either use a regular power supply or a 3 cell LiPo battery to provide the required voltage. Power sources can be connected to the relay ports on the Edge Control board, ensure that you have a source

for both the valve and the board.
Connect two jumper wires to the **GND** and **B** pins of the **Relay** ports.



Connect the jumper from the **B** pin to the positive terminal of the battery and the jumper from the **GND** pin to the negative terminal of the battery.

4. Uploading the Sketch

Connect the board to your computer, upload the `ValveControl.ino` sketch and open the **Serial Monitor**. If all the connections are done right, the valve opens and closes and you should be able to see the status as `Open` or `Close` on the serial monitor.

Conclusion

In this tutorial you learned how a 3 wire valve works and the basic operations that the Edge Control board uses to control the valves. With this knowledge you can build irrigation systems that periodically control the valves, which can be installed in your fields.

Complete Sketch

```
1  #include <Arduino_Edge
2
3  void setup(){
4      Serial.begin(9600)
5      while(!Serial);
6
7      delay(1000);
8
9      Serial.println("3-
10
11      EdgeControl.begin(
12      Latching.begin();
13
14      Serial.println("St
15  }
16
17  void loop(){
18      Serial.println("Cl
19      Latching.channelDi
20      Latching.strobe(45
21      delay(2500);
22
23      Serial.println("Op
24      Latching.channelDi
25      Latching.strobe(45
26      delay(2500);
27  }
```

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